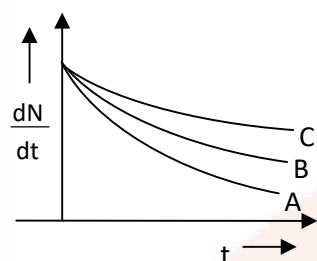


KCET – Physics – 2019

VERSION CODE: C-1

Question 1: Which one of the following nuclei has shorter mean life?



- (A) A
- (B) B
- (C) C
- (D) Same for all

Answer 1: (A)

Question 2: The conductivity of semiconductor increases with increase in temperature because

- (A) Number density of charge carriers increases
- (B) Relaxation time increases
- (C) Both number density of charge carriers and relaxation time increase
- (D) Number density of current carriers increases, relaxation time decreases but effect of decrease in relaxation time is much less than increase in number density

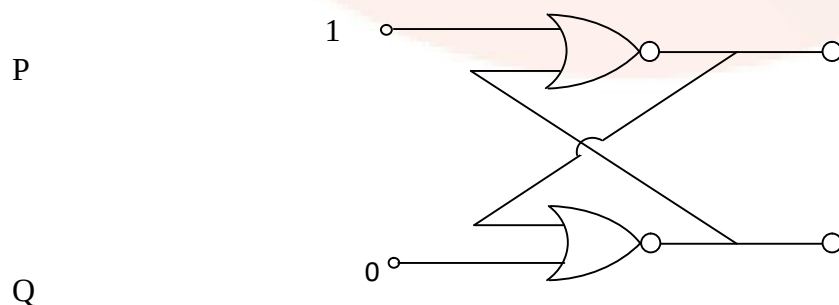
Answer 2: (D)

Question 3: For a transistor amplifier, the voltage gain

- (A) Remains constant for all frequencies
- (B) is high at high and low frequencies and constant in the middle frequency range
- (C) is low at high and low frequencies and constant at mid frequencies
- (D) Constant at high frequencies and low at low frequencies

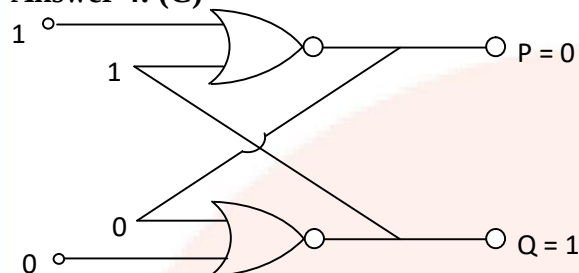
Answer 3: (C)

Question 4: In the following circuit, what are P and Q?



- (A) $P = 0, Q = 0$
- (B) $P = 1, Q = 0$
- (C) $P = 0, Q = 1$
- (D) $P = 1, Q = 1$

Answer 4: (C)



Question 5: An antenna uses electromagnetic waves of frequency 5 MHz. For proper working, the size of the antenna should be

- (A) 15 m
- (B) 300 m
- (C) 15 km
- (D) 3 km

Answer 5: (A)

Question 6: A magnetic needle has a magnetic moment of $5 \times 10^{-2} \text{ Am}^2$ and moment of inertia $8 \times 10^{-6} \text{ kgm}^2$. It has a period of oscillation of 2s in a magnetic field B . The magnitude of magnetic field is approximately?

- (A) $1.6 \times 10^{-4} \text{ T}$
- (B) $0.4 \times 10^{-4} \text{ T}$
- (C) $3.2 \times 10^{-4} \text{ T}$
- (D) $0.8 \times 10^{-4} \text{ T}$

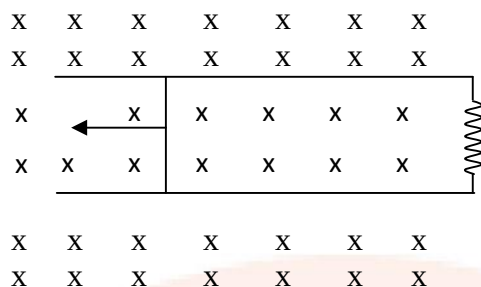
Answer 6: None of the options are correct.

Question 7: A toroid has 500 turns per metre length. If it carries a current of 2 A, the magnetic energy density inside the toroid is

- (A) 0.628 J/m^3
- (B) 0.314 J/m^3
- (C) 6.28 J/m^3
- (D) 3.14 J/m^3

Answer 7: (A)

Question 8: Consider the situation given in figure. The wire AB is slid on the fixed rails with a constant velocity. If the wire AB is replaced by a semicircular wire, the magnitude of the induced current will



- (A) Increases
(B) Remain same
(C) Decreases
(D) Increase or decrease depending on whether the semicircle bulges towards the resistance or away from it.

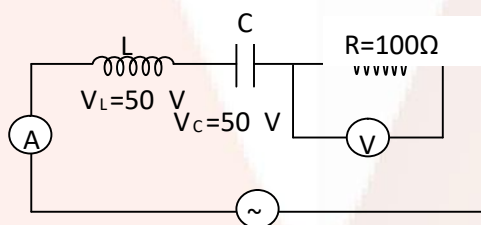
Answer 8: (B)

Question 9: The frequency of an alternating current is 50 Hz. What is the minimum time taken by current to reach its peak value from rms value?

- (A) 5×10^{-3} s
(B) 2.5×10^{-3} s
(C) 0.02s
(D) 10×10^{-3} s

Answer 9: (B)

Question 10: The readings of ammeter and voltmeter in the following circuit are respectively



- (A) 1.2 A, 120 V
(B) 1.5 A, 100 V
(C) 2.7 A, 220 V
(D) 2.2 A, 220 V

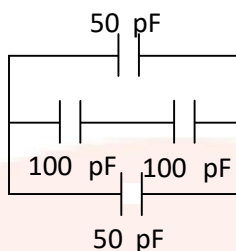
Answer 10: (C)

Question 11: Two metal plates are separated by 2 cm. The potentials of the plates are -10 V and +30 V. The electric field between the two plates is

- (A) 500 V/m
(B) 1000 V/m
(C) 2000 V/m
(D) 3000 V/m

Answer 11: (C)

Question 12: The equivalent capacitance between A and B is



- (A) 50 pF
- (B) $\frac{100}{3}$ pF
- (C) 150 pF
- (D) 300 pF

Answer 12: (B)

Question 13: A capacitor of capacitance C charged by an amount Q is connected in parallel with an uncharged capacitor of capacitance $2C$. The final charges on the capacitors are

- (A) $\frac{Q}{2}, \frac{Q}{2}$
- (B) $\frac{Q}{4}, \frac{3Q}{4}$
- (C) $\frac{Q}{3}, \frac{2Q}{3}$
- (D) $\frac{Q}{5}, \frac{4Q}{5}$

Answer 13: (C)

Question 14: Though the electron drift velocity is small and electron charge is very small, a conductor can carry an appreciably large current because

- (A) Electron number density is very large
- (B) Drift velocity of electron is very large
- (C) Electron number density depends on temperature
- (D) Relaxation time is small

Answer 14: (A)

Question 15: Masses of three wires of copper are in the ratio $1 : 3 : 5$ and their lengths are in the ratio $5 : 3 : 1$. The ratio of their electrical resistance are

- (A) $1 : 3 : 5$
- (B) $5 : 3 : 1$
- (C) $1 : 15 : 125$

(D) 125 : 15 : 1

Answer 15: (D)

Question 16: If P, Q and R are physical quantities having different dimensions, which of the following combinations can never be a meaningful quantity?

(A) $\frac{P-Q}{R}$

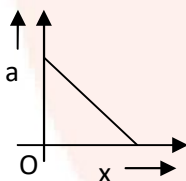
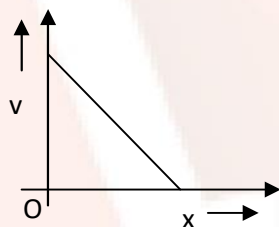
(B) $PQ - R$

(C) $\frac{PQ}{R}$

(D) $\frac{PR - Q^2}{R}$

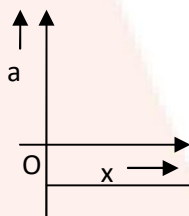
Answer 16: (A)

Question 17: The given graph shows the variation of velocity (v) with position (x) for a particle moving along a straight line

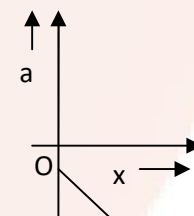
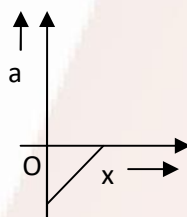


(A)

(C)



(D)



(B)

Answer 17: (C)

Question 18: The trajectory of a projectile projected from origin is given by the equation

$y = x - \frac{2x^2}{5}$. The initial velocity of the projectile is

(A) $\frac{2}{5} \text{ ms}^{-1}$

(B) 5 ms^{-1}

(C) 25 ms^{-1}

(D) $\frac{5}{2} \text{ ms}^{-1}$

Answer 18: (B)

Question 19: An object with mass 5 kg is acted upon by a force, $\vec{F} = (-3\hat{i} + 4\hat{j}) \text{ N}$. If its initial velocity at $t = 0$ is $\vec{v} = (3\hat{i} + 12\hat{j}) \text{ m/s}$, the time at which it will just have a velocity along y – axis is

- (A) 5 s
- (B) 10 s
- (C) 2 s
- (D) 15 s

Answer 19: (B)

Question 20: During inelastic collision between two objects, which of the following quantity always remains conserved?

- (A) Total kinetic energy
- (B) Total mechanical energy
- (C) Total linear momentum
- (D) Speed of each body

Answer 20: (C)

Question 21: In Rutherford experiment, for head-on collision of α -particles with a gold nucleus, the impact parameter is

- (A) Zero
- (B) of the order of 10^{-14} m
- (C) of the order 10^{-10} m
- (D) of the order of 10^{-6} m

Answer 21: (B)

Question 22: Frequency of revolution of an electron revolving in n^{th} orbit of H-atom is proportional to

- (A) $\frac{1}{n^2}$
- (B) n
- (C) n is independent of n
- (D) $\frac{1}{n^3}$

Answer 22: (B)

Question 23: A hydrogen atom in ground state absorbs 10.2 eV of energy. The orbital angular momentum of the electron is increased by

- (A) $1.05 \times 10^{-34} \text{ Js}$
- (B) $2.11 \times 10^{-34} \text{ Js}$

- (C) $3.16 \times 10^{-34} \text{Js}$
(D) $4.22 \times 10^{-34} \text{Js}$

Answer 23: (A)

Question 24: The end product of decay of ${}_{90}\text{Th}^{32}$ is ${}_{82}\text{Pb}^{208}$. The number of α and β particles emitted are respectively

- (A) 3, 3
(B) 6, 4
(C) 6, 0
(D) 4, 6

Answer 24: (B)

Question 25: Two protons are kept at a separation of 10nm. Let F_n and F_e be the nuclear force and the electromagnetic force between them

- (A) $F_e = F_n$
(B) $F_e \gg F_n$
(C) $F_e \ll F_n$
(D) F_e and F_n differ only slightly

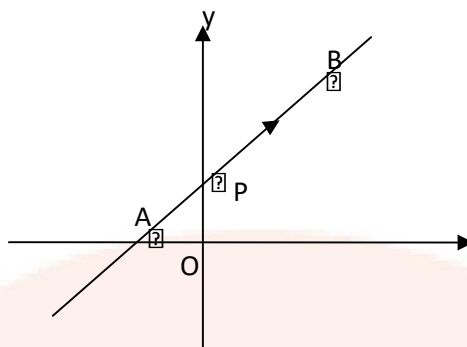
Answer 25: (B)

Question 26: Two particles which are initially at rest move towards each other under the action of their mutual attraction. If their speeds are v and $2v$ at any instant, then the speed of centre of mass of the system is

- (A) $2v$
(B) Zero
(C) $1.5v$
(D) v

Answer 26: (B)

Question 27: A particle is moving uniformly along a straight line as shown in the figure. During the motion of the particle from A to B, the angular momentum of the particle about 'O'



- (A) Increase
- (B) Decrease
- (C) Remain constant
- (D) First increase then decrease.

Answer 27: (C)

Question 28: A satellite is orbiting close to the earth and has a kinetic energy K . The minimum extra kinetic energy required by it just overcome the gravitation pull of the earth is

- (A) K
- (B) $2K$
- (C) $\sqrt{3}$
- (D) $2\sqrt{2}$

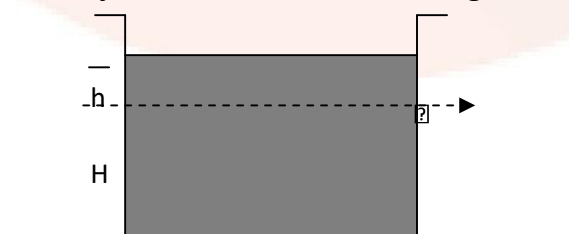
Answer 28: (A)

Question 29: A wire is stretched such that its volume remains constant. The poisson's ratio of the material of the wire is

- (A) 0.50
- (B) -0.50
- (C) 0.25
- (D) -0.25

Answer 29: (B)

Question 30: A cylindrical container containing water has a small hole at height of $H = 8$ cm from the bottom and at a depth of 2cm from the top surface of the liquid. The maximum horizontal distance travelled by the water before it hits the ground (x) is



(A) 8cm

(B) $4\sqrt{2}$ cm

(C) 4cm

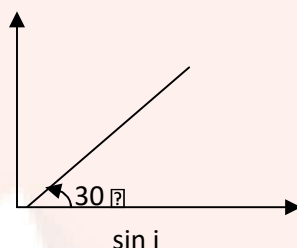
(B) $4\sqrt{2}$

(C) 4 cm

(D) 6 cm

Answer 30: (A)

Question 31: A transparent medium shows relation between I and r as shown. If the speed of light in vacuum is c the Brewster angle for the medium is



(A) 30°

(B) 45°

(C) 60°

(D) 90°

Answer 31: (B)

Question 32: In Young's double slit experiment, using monochromatic light of wavelength λ , the intensity of light at a point on the screen where path difference is λ is K units. The intensity of light at

(A) K

(B) $\frac{K}{4}$

(C) $4K$

(D) $2K$

Answer 32: (B)

Question 33: Due to Doppler's effect the shift in wavelength observed is $0.1\text{\AA}$ for a star producing wavelength $6000\text{\AA}$. Velocity of recession of the star will be

(A) 25 km/s

(B) 10 km/s

(C) 5 km/s

(D) 20 km/s

Answer 33: (C)

Question 34: Now-a-days agricultural practice is expensive to the farmers as they need to purchase hybrid seeds every year. Which of the following strategies can be employed to overcome this problem?

(A) Remains constant

- (B) Increases with time
- (C) Decrease with time
- (D) Increase and decrease periodically

Answer 34: (A)

Question 35: Light of certain frequency and intensity incident on a photosensitive material causes photoelectric effect. If both the frequency and intensity are doubled, the photoelectric saturation current becomes.

- (A) Quadruple
- (B) Doubled
- (C) Halved
- (D) Unchanged

Answer 35: (B)

Question 36: A certain charge $2Q$ is divided at first into two parts q_1 and q_2 . Later the charges are placed at a certain distance. If the force of interaction between two charges is maximum then

- (A) 4
- (B) 2
- (C) 1
- (D) 0.5

Answer 36: (C)

Question 37: A particle of mass m and charge q is placed at rest in uniform electric field E and then released. The kinetic energy attained by the particle after moving a distance y is

- (A) qEy^2
- (B) qE^2y
- (C) qEy
- (D) q^2Ey

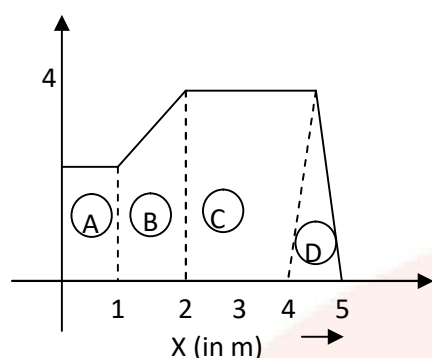
Answer 37: (C)

Question 38: An electric dipole is kept in non-uniform electric field. It generally experiences

- (A) A force and torque
- (B) A force but not a torque
- (C) A torque but not a force
- (D) Neither a force nor a torque

Answer 38: (A)

Question 39: The figure gives the electric potential V as a function of distance through four regions on x-axis. Which of the following is true for the magnitude of the electric field E in these regions?



- (A) $E_A > E_B > E_C > E_D$
 (B) $E_A = E_C$ and $E_B < E_D$
 (C) $E_B = E_D$ and $E_A < E_C$

(D) $E_A < E_B < E_C < E_D$

Answer 39: (B)

Question 40: A system of two charges separated by a certain distance apart stores electrical potential energy. If the distance between them is increased, the potential energy of the system.

- (A) $E_A > E_B > E_C > E_D$
 (B) decreases in any case
 (C) may increase or decrease
 (D) remains the same

Answer 40: (C)

Question 41: In a cyclotron a charged particle

- (A) undergoes acceleration all the time
 (B) speeds up between the dees because of the magnetic field
 (C) speeds up in dee
 (D) slows down within a dee and speeds up between dees

Answer 41: (A)

Question 42: The number of turns in a coil of Galvanometer is tripled, then

- (A) Voltage sensitivity increases 3 times and current sensitivity remains constant.
 (B) Voltage sensitivity remains constant and current sensitivity increases 3 times.
 (C) Both voltage and current sensitivity remains constant.
 (D) Both voltage and current sensitivity decreases by 33%

Answer 42: (B)

Question 43: A circular current loop of magnetic moment M is in a arbitrary orientation in an external uniform magnetic field $\vec{B}$. The work done to rotate the loop by 30° about an axis perpendicular to its plane is

- (A) MB
 (B) $\sqrt{3} \frac{MB}{2}$

(C) $\frac{MB}{2}$

(D) Zero

Answer 43: (D)

Question 44: In a permanent magnet at room temperature

(A) Magnetic moment of each molecule is zero

(B) The individual molecules have non zero magnetic moment which are all perfectly aligned.

(C) Domains are partially aligned

(D) Domains are all perfectly aligned

Answer 44 : (C)

Question 45: Coersivity of a magnet where the ferromagnet gets completely demagnetized is $3 \times 10^3 \text{ Am}^{-1}$. The minimum current required to be passed in a solenoid having 1000 turns per metre, so that the magnet gets completely demagnetized when placed inside the solenoid is

(A) 30mA

(B) 60mA

(C) 3A

(D) 6A

Answer 45: (C)

Question 46: An inductor of inductance L and resistor R are joined together in series and connected by a source of frequency ω . The power dissipated in the circuit is

(A) $\frac{R^2 + \omega^2 L^2}{V}$

(B) $\frac{V^2 R}{R^2 + \omega^2 L^2}$

(C) $\frac{V}{R^2 + \omega^2 L^2}$

(D) $\frac{V^2 R}{\sqrt{R^2 + \omega^2 L^2}}$

Answer 46: (B)

Question 47: An electromagnetic wave is travelling in x-direction with electric field vector given by, $\vec{E}_y = E_0 \sin(kx - \omega t) \hat{j}$. The correct expression for magnetic vector is

(A) $\vec{B}_y = E_0 C \sin(kx - \omega t) \hat{j}$

(B) $\vec{B}_z = E_0 C \sin(kx - \omega t) \hat{k}$

(C) $\vec{B}_y = \frac{E_0}{C} \sin(kx - \omega t) \hat{j}$

(D) $\vec{B}_z = \frac{E_0}{C} \sin(kx - \omega t) \hat{k}$

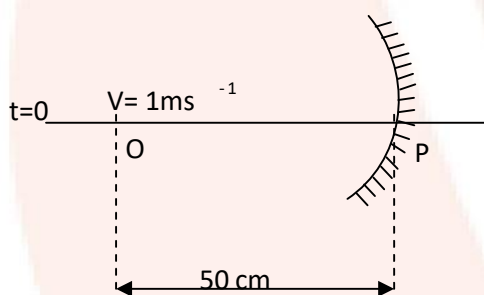
Answer 47: (D)

Question 48: The phenomenon involved in the reflection of radio-waves by ionosphere is similar to

- (A) Reflection of light by plane mirror
- (B) Total internal reflection of light in air during a mirage
- (C) Dispersion of light by water molecules during the formation
- (D) Scattering of light by air particles

Answer 48: (B)

Question 49: A point object is moving uniformly towards the pole of a concave mirror of focal length 25cm along its axis as shown below. The speed of the object is 1 ms^{-1} . At $t = 0$, the distance of the object from the mirror is 50cm. the average velocity of the image formed by the mirror between time $t = 0$ and $t = 0.25 \text{ s}$ is:



(A) 40 cm s^{-1}

- (B) 420 cm s^{-1}
- (C) Zero
- (D) Infinity

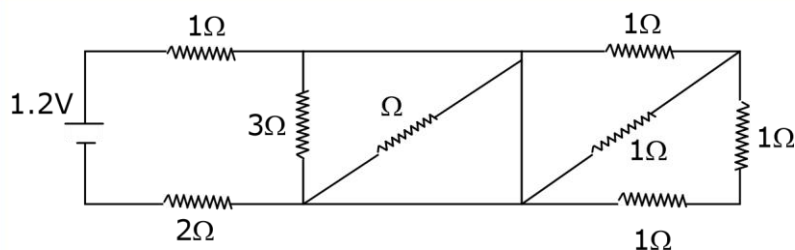
Answer 49: (D)

Question 50: A certain prism is found to produce a minimum deviation of 44° when the angle of incidence is either 42° or 62° . What is the angle of incidence when it is undergoing minimum deviation?

- (A) 30°
- (B) 40°
- (C) 49°
- (D) 60°

Answer 50: (C)

Question 51: In the given circuit, the current through 2Ω resistor is



- (A) 0.2A
- (B) 0.3A
- (C) 0.4A
- (D) 0.1A

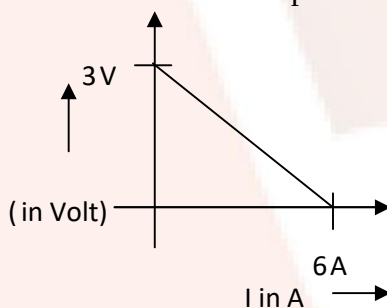
Answer 51: (C)

Question 52: Kirchhoff's junction rule is a reflection of

- (A) Conservation of current density vector
- (B) Conservation of energy
- (C) Conservation of momentum
- (D) Conservation of charges

Answer 52: (D)

Question 53: The variation of terminal potential difference (V) with current flowing through a cell is as shown:



- (A) 3V, 2Ω
- (B) 3V, 0.5Ω
- (C) 6V, 2Ω
- (D) 6V, 0.5Ω

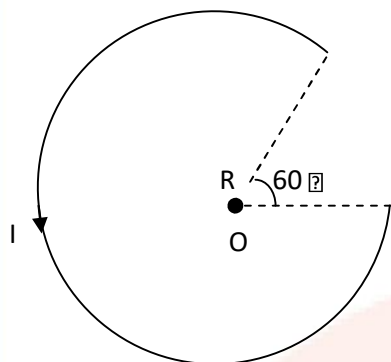
Answer 53: (B)

Question 54: In a potentiometer experiment, the balancing point with a cell is at a length 240 cm. on shunting the cell with a resistance of 2Ω , the balancing length becomes 120cm. The internal resistance of the cell is

- (A) 4Ω
- (B) 2Ω
- (C) 1Ω
- (D) 0.5Ω

Answer 54: (B)

Question 55: The magnetic fields at the centre 'O' in the given figure is



- (A) $\frac{7}{14} \frac{\mu_0 I}{R}$
 (B) $\frac{5}{12} \frac{\mu_0 I}{R}$
 (C) $\frac{3}{10} \frac{\mu_0 I}{R}$
 (D) $\frac{\mu_0 I}{12R}$

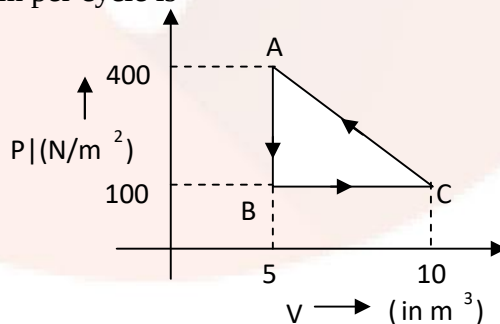
Answer 55: (B)

Question 56: An aluminium sphere is dipped into water. Which of the following is true?

- (A) Buoyancy will be less in water at 0°C than that in water at 4°C
 (B) Buoyancy will be more in water at 0°C than that in water at 4°C
 (C) Buoyancy in water at 0°C will be same as that in water at 4°C
 (D) Buoyancy may be more or less in water at 4°C depending on the radius of the sphere

Answer 56: (A)

Question 57: A thermodynamic system undergoes a cyclic process ABC as shown in the diagram. The work done by the system per cycle is



- (A) 750J
 (B) -1250J
 (C) -750J
 (D) 1250J

Answer 57: (C)

Question 58: One mole of O_2 gas is heated at constant pressure starting at $27^\circ C$. How much energy must be added to the gas as heat to double its volume?

- (A) Zero
- (B) $450R$
- (C) $750R$
- (D) $1050R$

Answer 58: (D)

Question 59: A piston is performing S.H.M. in the vertical direction with a frequency of 0.5 Hz . A block of 10 kg is placed on the piston. The maximum amplitude of the system such that the block remains in contact with the piston is

- (A) 1m
- (B) 0.5m
- (C) 1.5m
- (D) 0.1m

Answer 59: (A)

Question 60: The equation of a stationary wave is $y = 2 \sin\left(\frac{\pi X}{15}\right) \cos(48\pi t)$. The distance between a node and its next antinode is

- (A) 7.5 units
- (B) 1.5 units
- (C) 22.5 units
- (D) 30 units

Answer 60: (A)